



GENERATIVE AI BASED BEHAVIOURAL INTERVIEW PREPARATION AND GUIDANCE SYSTEM IN IT SECTOR

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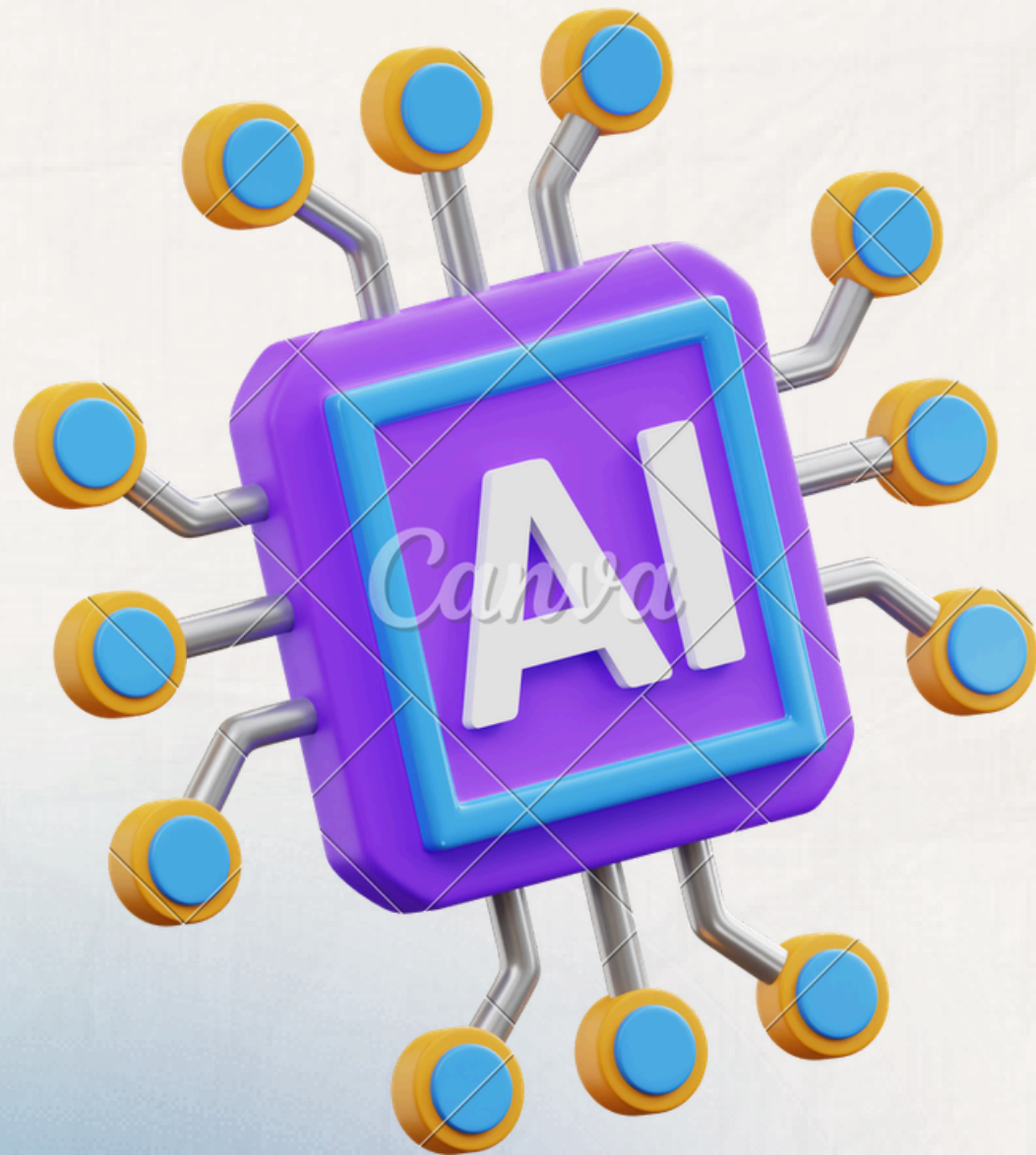
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INTRODUCTION

- An AI-based behavioural interview preparation system designed for students, graduates, and job seekers in the IT sector.
- Creates a personalized profile using users' career goals, interests, domain, and experience level.
- Provides domain- and level-specific behavioural interview questions through an AI chat assistant.
- Supports both practice and mock interview modes using text-based interview sessions.
- Evaluates users' responses using AI/NLP techniques and provides personalized feedback on strengths and areas for improvement.
- Helps users improve interview readiness, confidence, and behavioural interview performance through automated practice and evaluation.

OBJECTIVE

- To develop a Generative AI-Based Behavioural Interview Preparation and Guidance System that provides users with conversational AI guidance, personalized mock behavioural interview practice, and automated performance evaluation through explainable scoring, feedback, and progress reports.

MOTIVATION

- Increasing competition makes behavioural interview preparation increasingly important.
- Many candidates struggle with nervousness, incomplete answers, and presenting their skills effectively.
- Existing platforms mainly provide static questions with limited and non-personalized feedback.
- There is a need for an intelligent and personalized system that combines:
 - Conversational AI guidance
 - Mock behavioural interviews
 - Automated performance evaluation
 - Personalized and explainable feedback
- The system aims to provide adaptive interview preparation in a single platform.

PROBLEM STATEMENT



01.

Existing platforms provide general AI guidance but lack structured behavioural mock interview practice.

02.

Many interview systems focus on scoring responses rather than supporting continuous practice and guidance.

03.

Users often cannot manage personalized profiles or track their scores, feedback, and improvement over time.

04.

Existing systems may provide unclear scoring and limited explanations, reducing user trust and confidence.

SCOPE AND APPLICATION

The system is designed for students, graduates, and job seekers preparing for interviews. It provides career guidance, mock interview sessions, and automated feedback to help users improve their performance.

Applications:

- Users register or log in and build a profile with their skills and interests.
- Users interact with a Local AI/LLM agent through open chat for general guidance.
- Users participate in structured, AI-generated mock behavioural interview sessions.
- Responses are evaluated by the AI agent, generating scores, feedback, and progress reports.
- An administrative interface manages users, behavioural interview questions, and system reports.

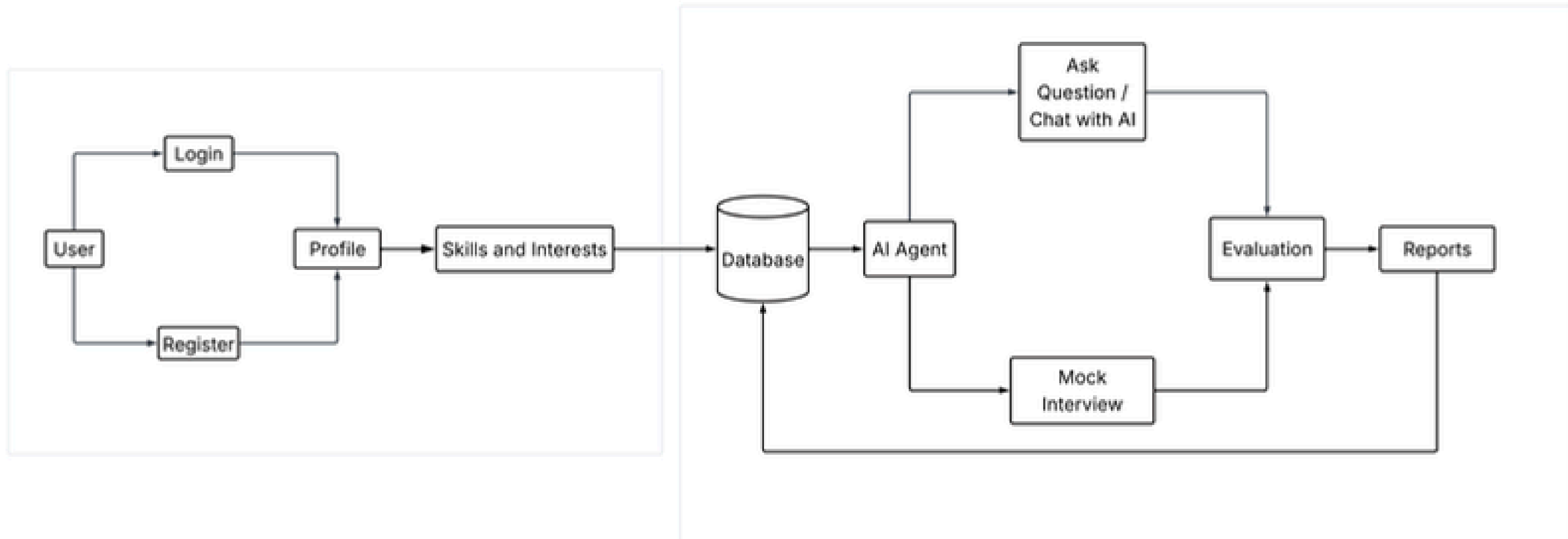
METHODOLOGY

This section presents the overall methodology adopted for the development of the Generative AI Based Behavioural Interview Preparation And Guidance System In IT Sector.

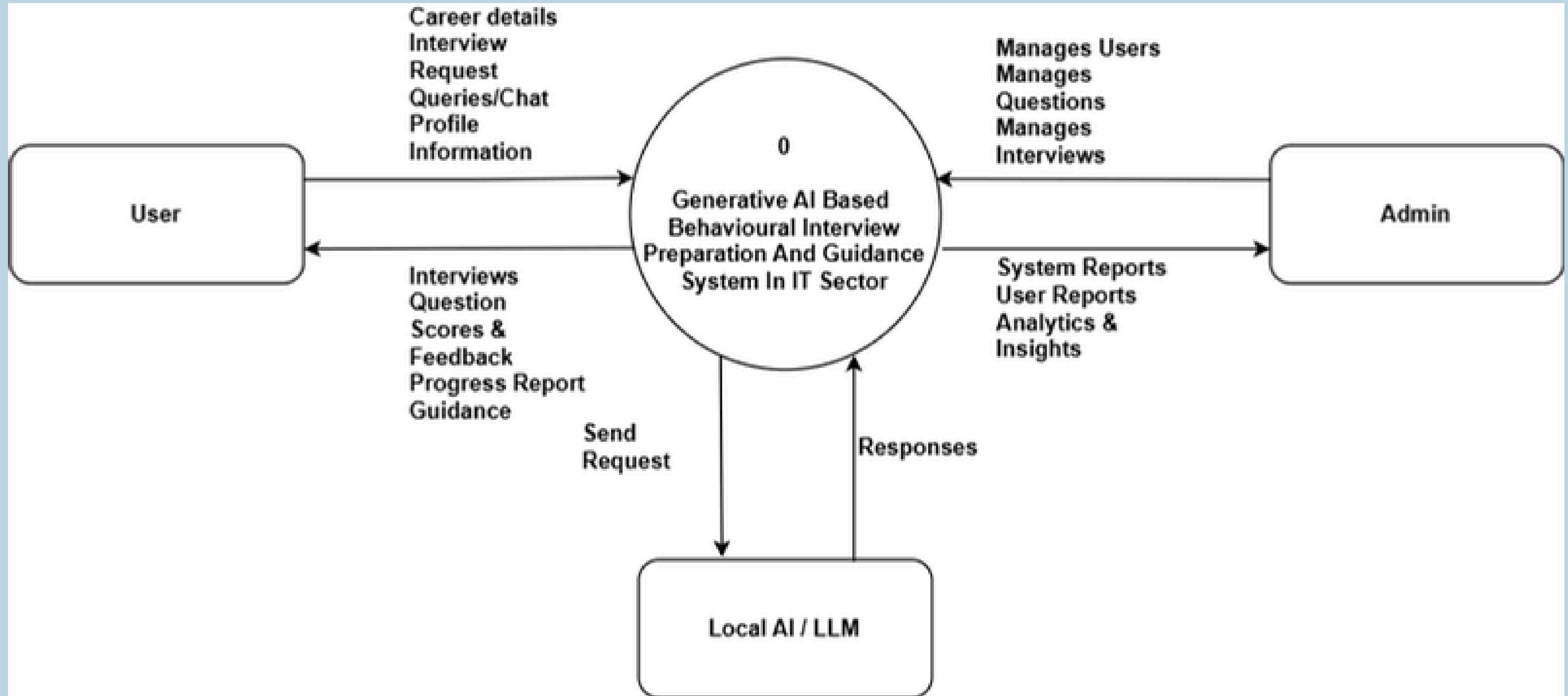
Technology Stack

- Ollama + Llama 3.2: Generates personalized behavioural interview questions and reference answers based on role, difficulty, and practice history.
- Groq API: Provides real-time AI chat, interview guidance, and personalized career advice using the user's profile and conversation history.
- Nomic-Embed-Text: Converts reference and user answers into 384-dimensional embeddings and evaluates their semantic similarity using cosine similarity.
- Local Processing: Question generation and response evaluation run locally, providing privacy, lower cost, and reduced latency.

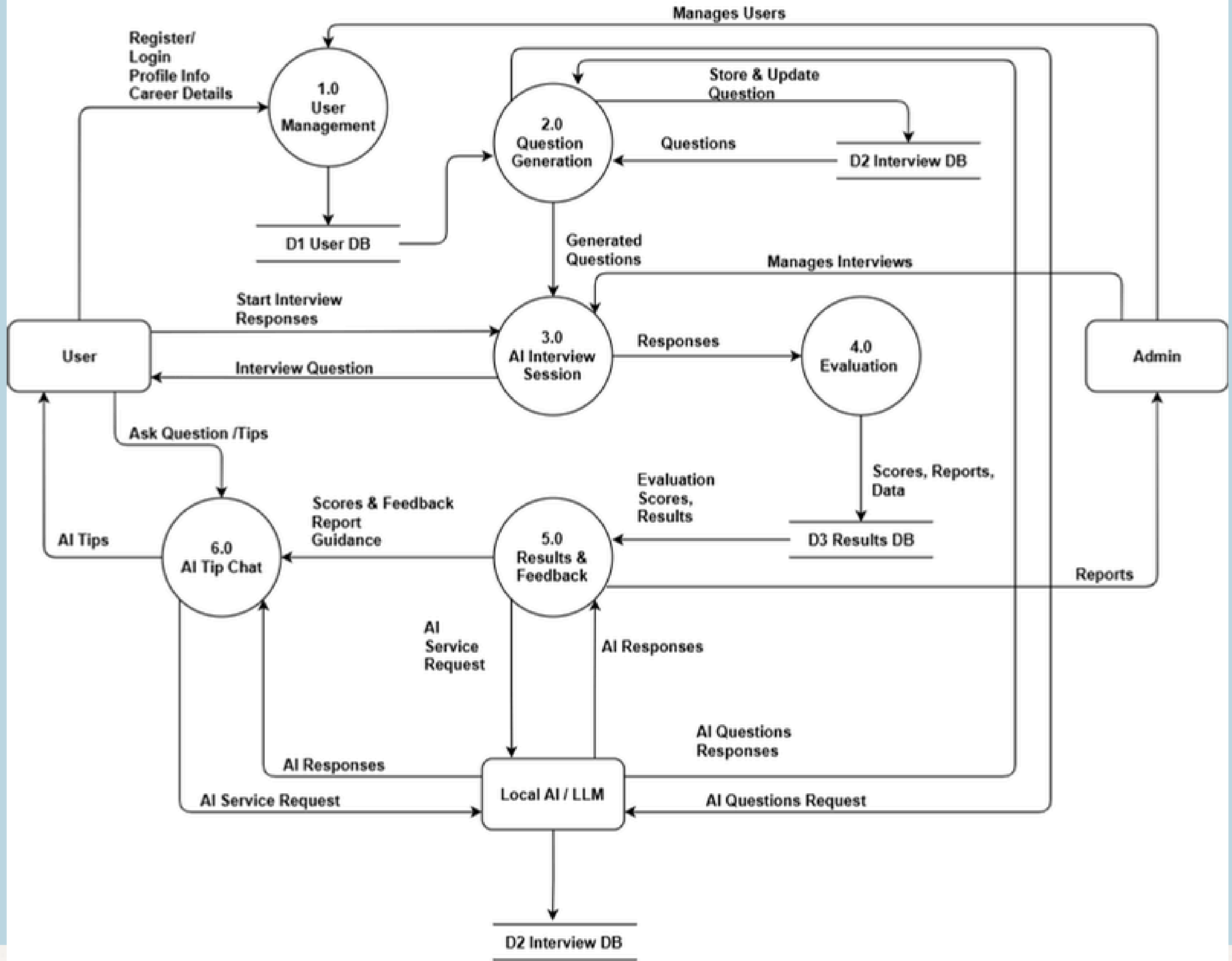
BLOCK DIAGRAM



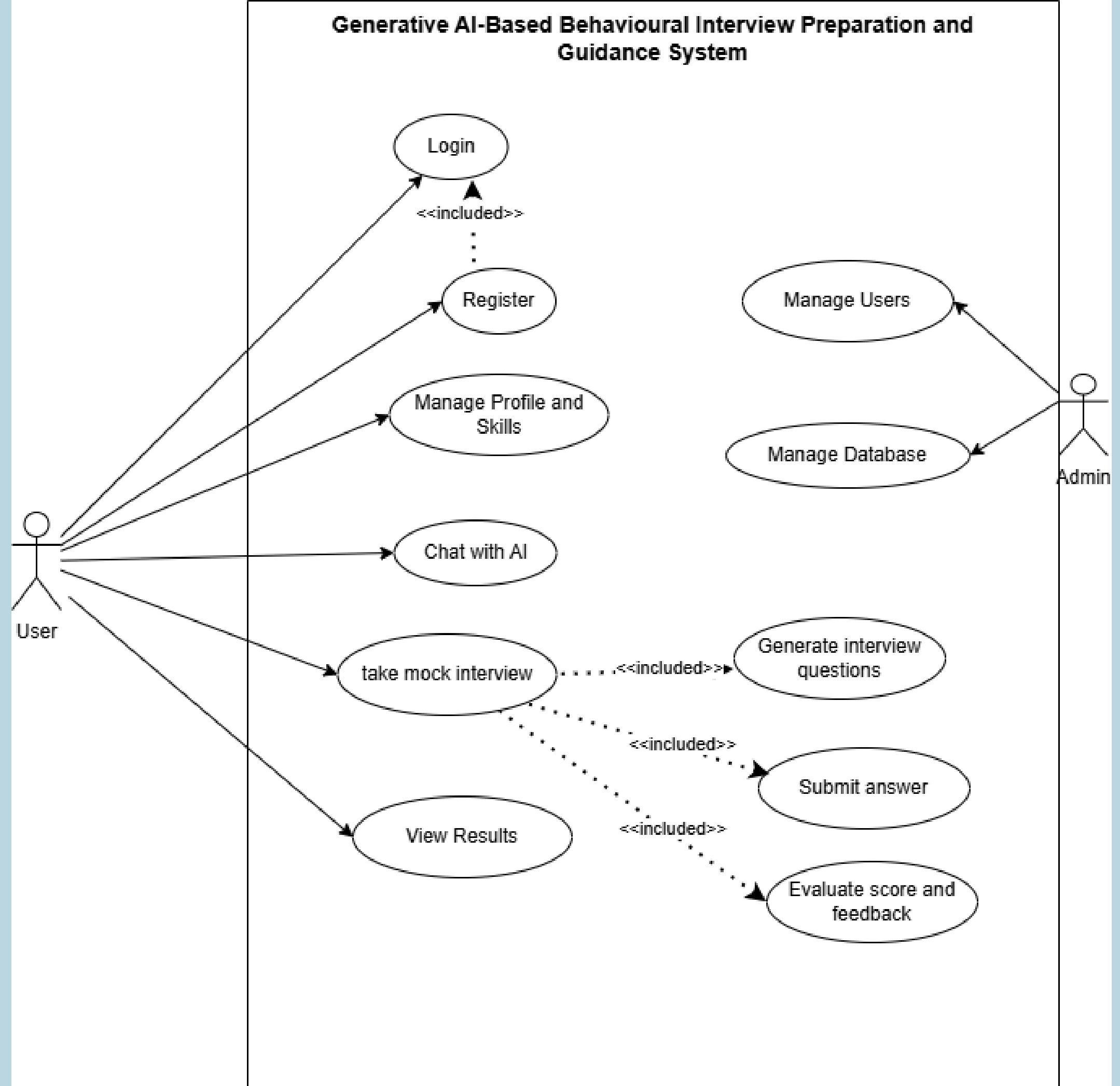
DFD LEVEL 0



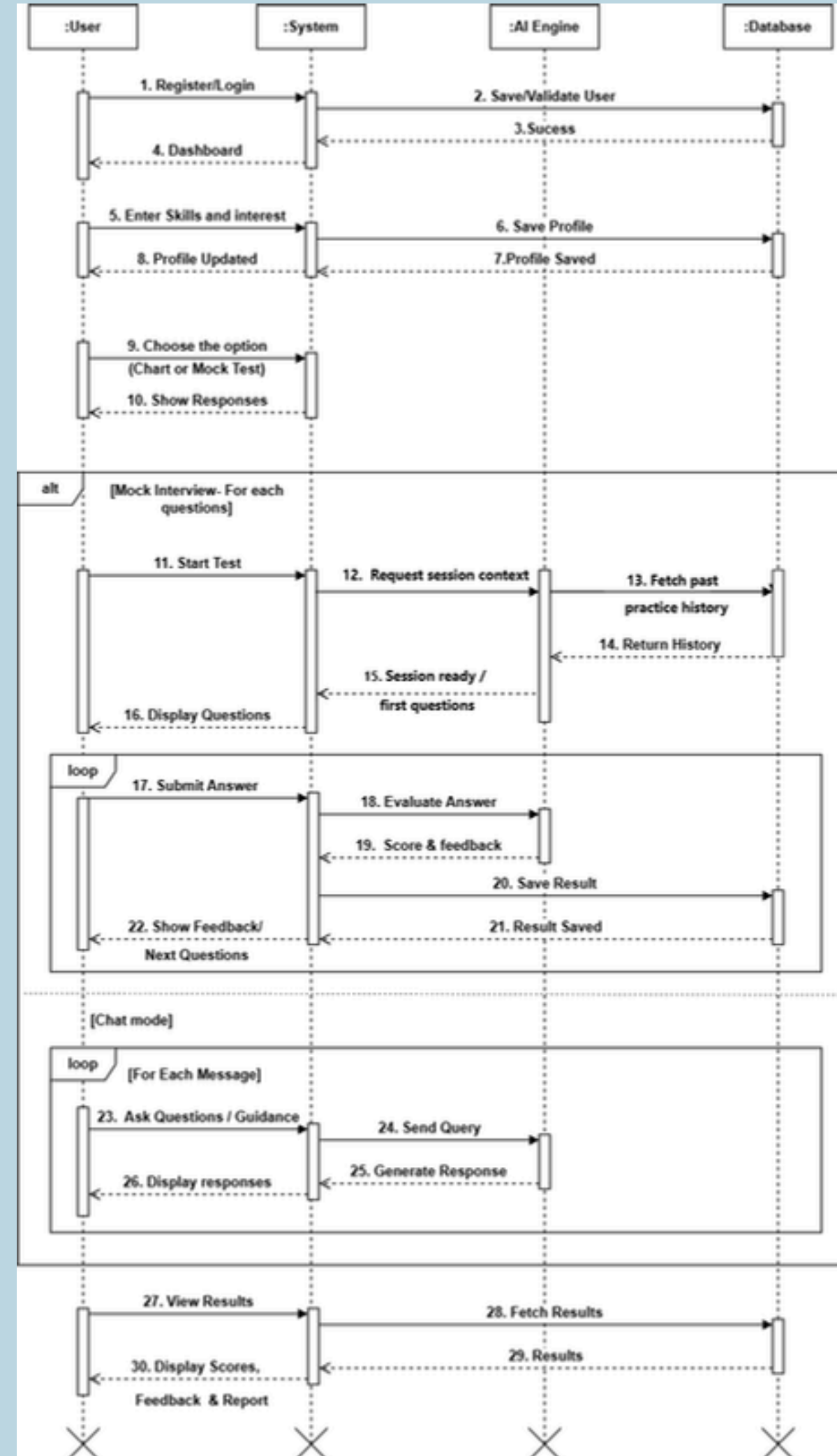
DFD LEVEL 1



USECASE DIAGRAM



SEQUENCE DIAGRAM



FEASIBILITY STUDY

Technical Feasibility

- Uses mature technologies: LLMs, NLP, Ollama, Groq API, and Nomic-Embed-Text. Built with Django REST Framework, React.js, and PostgreSQL.
- Requires standard computing resources and does not require model retraining.

Operational Feasibility

- Provides a simple and user-friendly platform for students and job seekers.
- Integrates profile setup, AI chat guidance, mock interviews, and automated evaluation.
- Provides scores, feedback, and progress tracking to improve interview performance.

Scheduling Feasibility

- Can be completed within the allocated project duration through proper planning.
- Covers requirement analysis, design, development, testing, and documentation.
- Regular monitoring and reviews ensure timely completion of project milestones.

DATA COLLECTION & DATASETS

- Easy Level: 320 questions from Kaggle's HR Interview Questions and Ideal Answers, focusing on basic behavioural skills, communication, and work ethic.
- Medium Level: 150 behavioural and situational questions manually selected from Exponent, focusing on problem-solving, decision-making, and strategic thinking.
- Hard Level: 20,444 advanced behavioural interview questions from Hugging Face's InterviewForge_GenDS, covering complex scenarios, leadership, and strategic decision-making.
- The three datasets provide questions across Easy, Medium, and Hard difficulty levels to support the system's adaptive interview generation.

Data Processing

- Format Conversion – Converted all three datasets into a unified Excel (.xlsx) format.
- Data Cleaning – Removed unnecessary columns, corrected missing values and formatting issues, and cleaned malformed entries.
- Duplicate Removal – Removed duplicate question–answer pairs to reduce redundancy and bias.
- Difficulty Labeling – Classified questions as Easy, Medium, or Hard based on their content.
- Embedding Generation – Used Nomic-Embed-Text via Ollama to convert questions and reference answers into dense vector embeddings for semantic retrieval.

EVALUATION METRICS

Retrieval Evaluation

- Similarity Score: $1 - \text{Distance}$; average similarity of the top 5 retrieved questions. Value: 0.685
- Precision@5: $\text{Relevant questions in Top 5} \div 5$. Value: 0.252
- Recall@5: $\text{Relevant question retrieved in Top 5} \div \text{Total relevant questions}$. Value: 0.365
- MRR: $\text{Average of } 1 \div \text{rank of the first relevant result}$. Value: 0.958
- F1 Score: $2 \times \text{Precision} \times \text{Recall} \div (\text{Precision} + \text{Recall})$. Value: 0.295

EVALUATION METRICS

Generation Evaluation

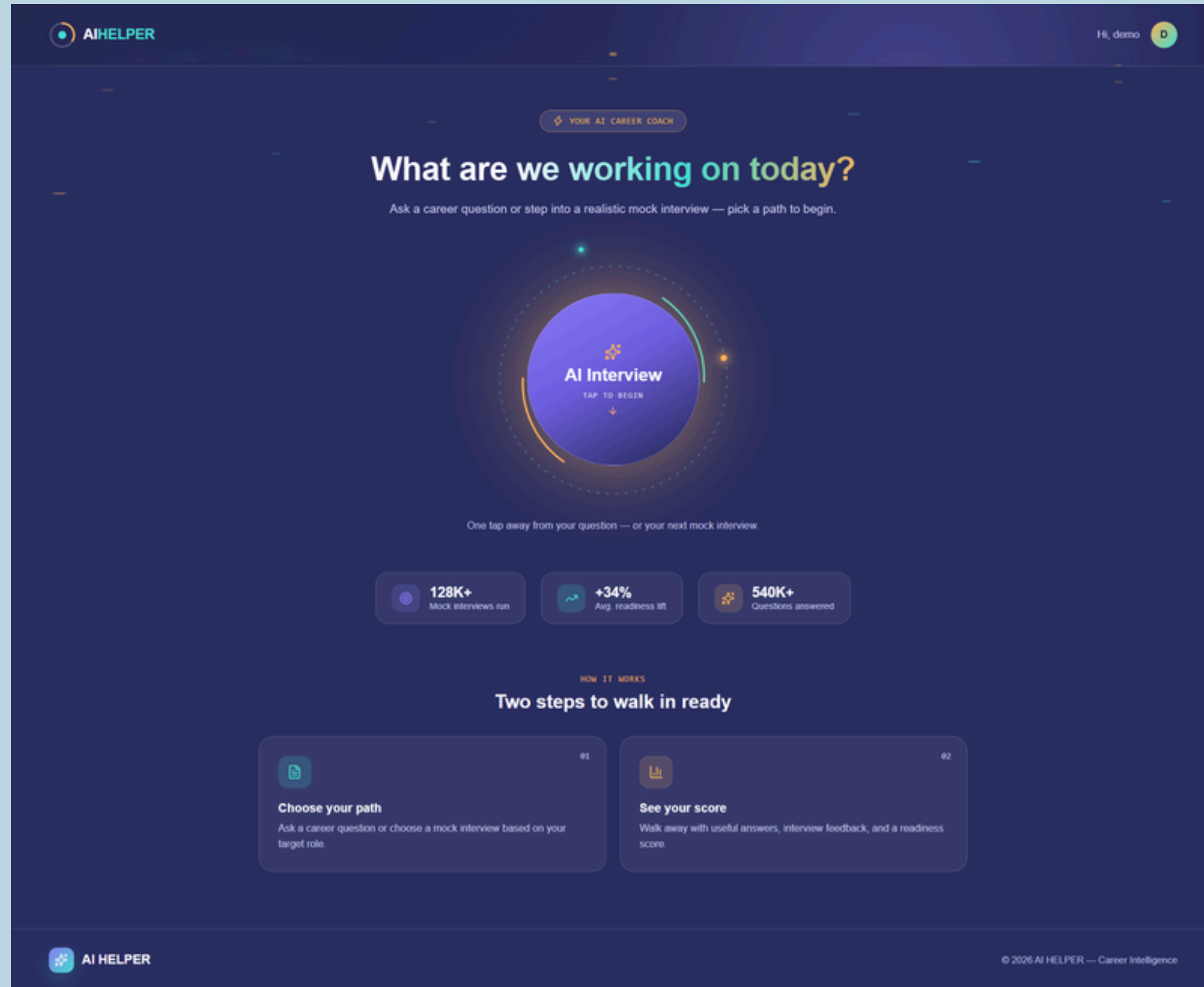
- Exact Match: Exactly matched responses ÷ Total responses. Value: 0.100
- BLEU: Measures matching n-grams between generated and reference answers with a brevity penalty. Value: 0.170
- ROUGE-1: F1 score based on individual word (unigram) overlap. Value: 0.345
- ROUGE-2: F1 score based on two-word sequence (bigram) overlap. Value: 0.210
- ROUGE-L: F1 score based on the Longest Common Subsequence (LCS). Value: 0.305

REQUIREMENTS

The system operates with the following software and hardware specifications:

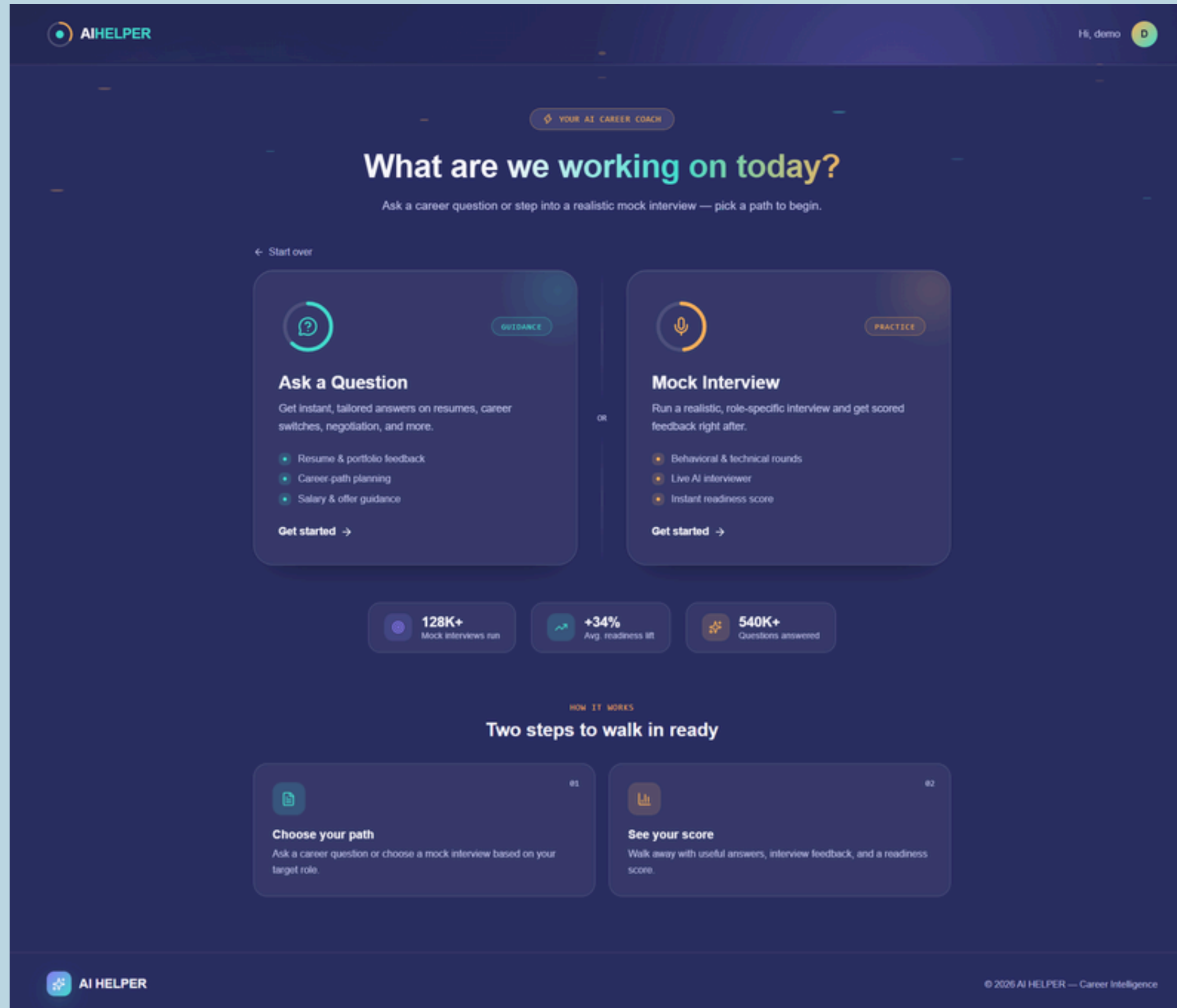
- Back-end & AI: Python, Django/Django REST Framework for backend development and Generative AI APIs/models.
- Front-end: React.js, HTML5, Tailwind CSS, and JavaScript for developing the interactive user interface.
- Database: PostgreSQL for storing user accounts, interview-related data, questions, answers, scores, and feedback.
- Development Tools: Visual Studio Code, Git/GitHub, Docker and Docker Compose for development, version control, and containerized deployment.
- Hardware: The system has been developed and tested on a standard computer with at least 8 GB RAM, a modern multi-core processor, and sufficient storage for datasets, dependencies, and application containers.

WORK DONE



HOME PAGE

WORK DONE



OPTION PAGE

WORK DONE

Create your account

Start your journey toward interview success.

Full Name

Email Address

Password

Confirm Password

☐ I agree to the Terms of Service and Privacy Policy.

Create Account

Already have an account? [Login](#)

Welcome back

Sign in to continue your interview preparation.

Email Address

Password [Forgot password?](#)

☐ Remember me

Login

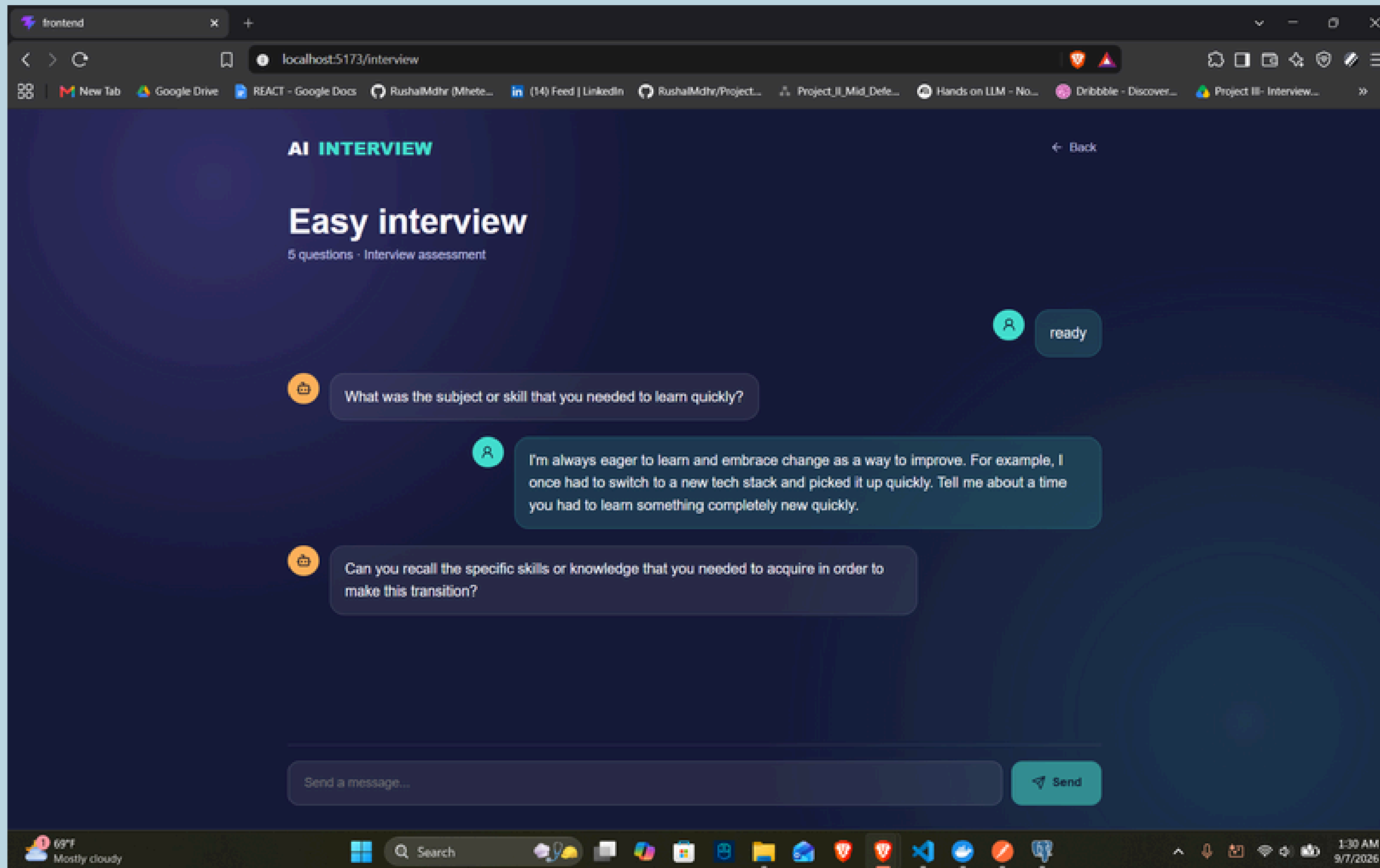
OR

Continue with Google

Don't have an account? [Create account](#)

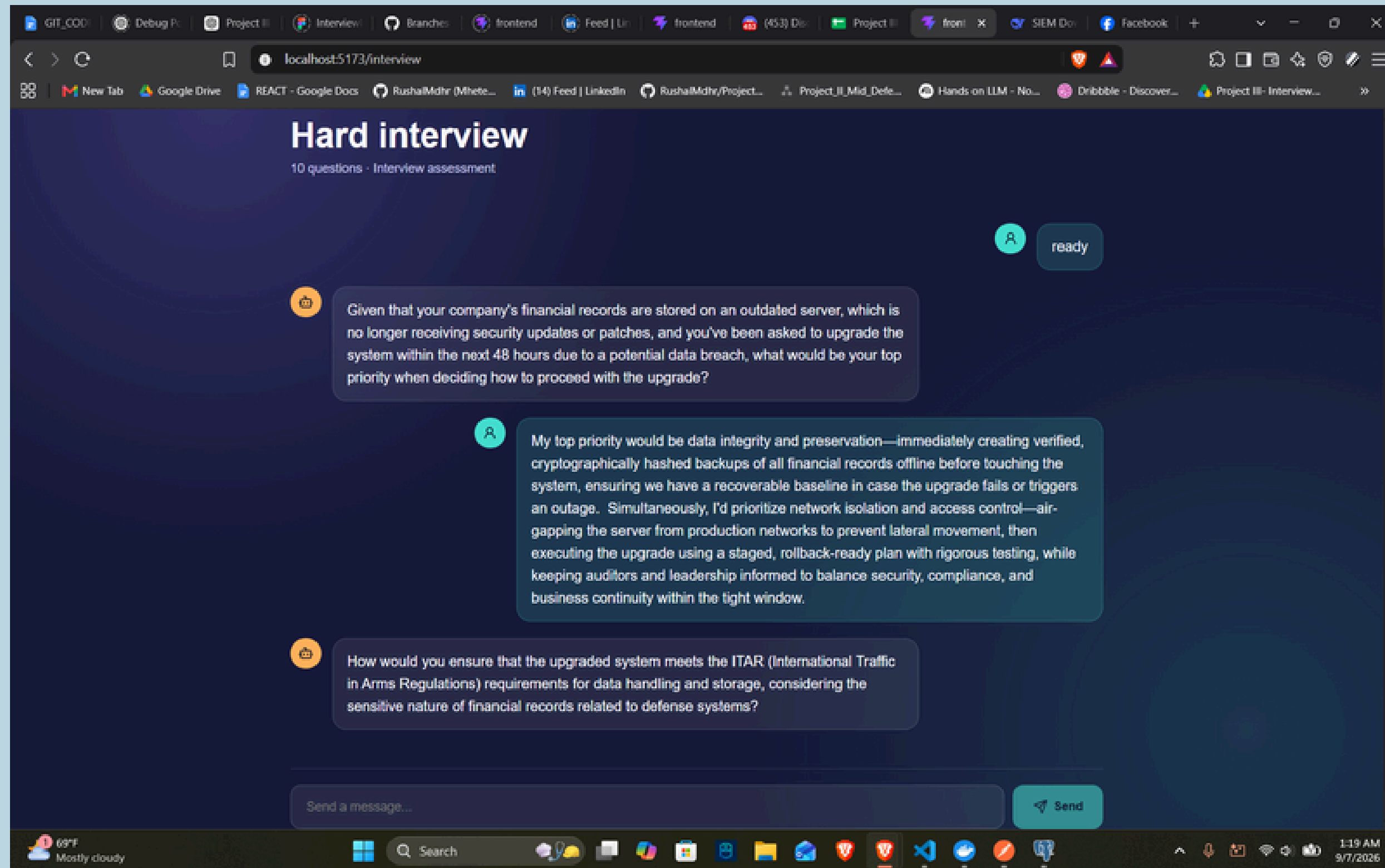
create account (Register and Login)t

WORK DONE



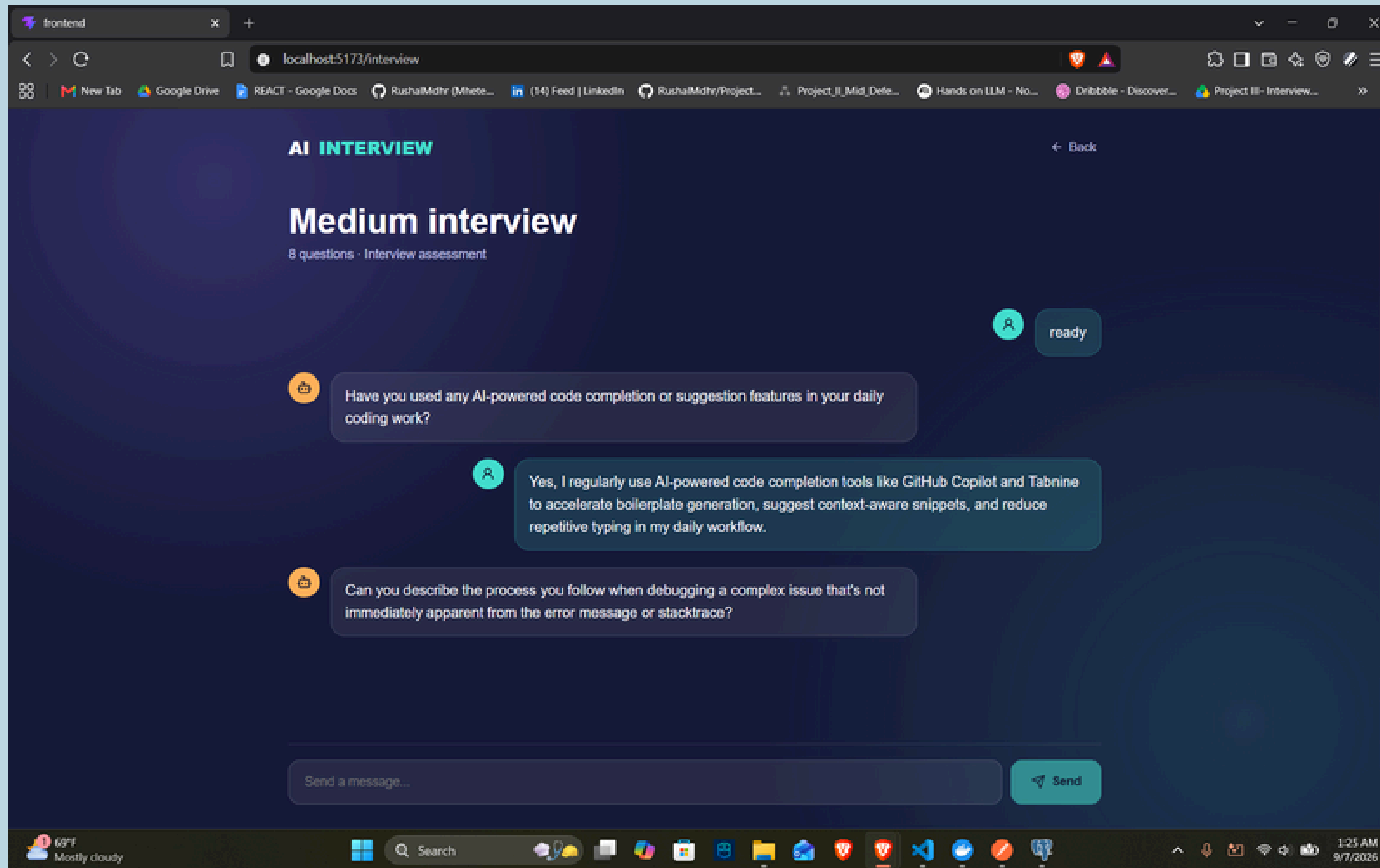
mock interview

WORK DONE



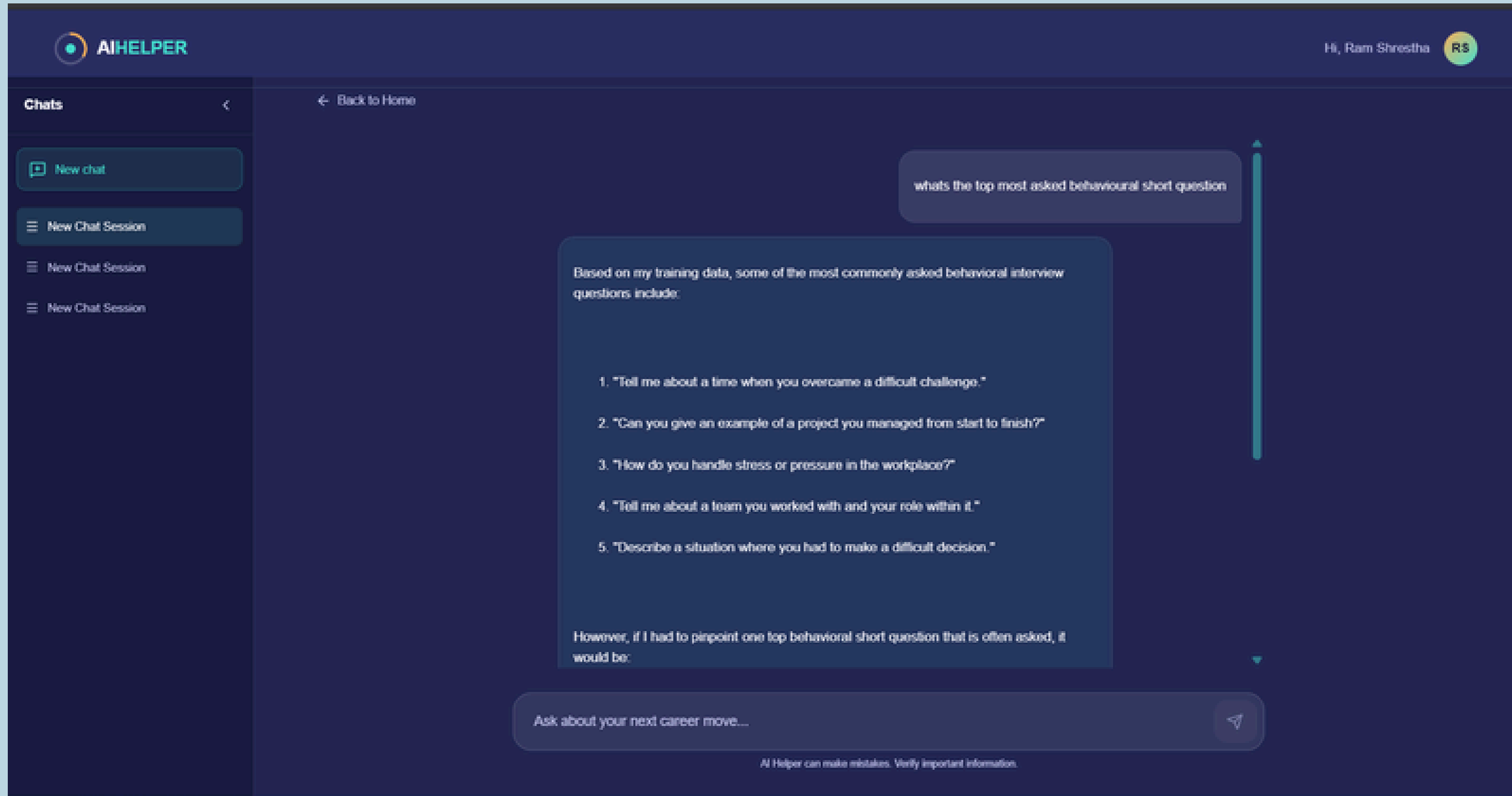
mock interview

WORK DONE



mock interview

WORK DONE



AI Chat

WORK REMAINING

- Users Profile and Dashboard
- Users interview Evaluation.

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THANK YOU
VERY MUCH